Device Programming Coursework
I^{2}C Modular Design Documentation

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1 Modular Device Overview

The device consists of 4 boxes:

- The Bus - is connected to a power supply and provides the 4 wire connections for the other boxes;
- The Controller - has two buttons that can pull the SDA and SCL lines low;
- The Listener - provides the state of the current byte being transmitted through LEDs;
- The Target - simulates 4 individual 'targets' that the controller can talk to;

All boxes are made of plywood. With the exception of the Controller, they have the same outside dimensions and differ only by the hole layout on the top face.

Connections between the boxes are made through 4 PIN DIN connectors. The Bus has 4 female connectors on the top face, and every other box has a male connector. The wire color-code is:

- brown - 5V line
- gray - Ground
- pink - SDA
- green - SCL (with the exception of the Listener Box, where SCL is red)

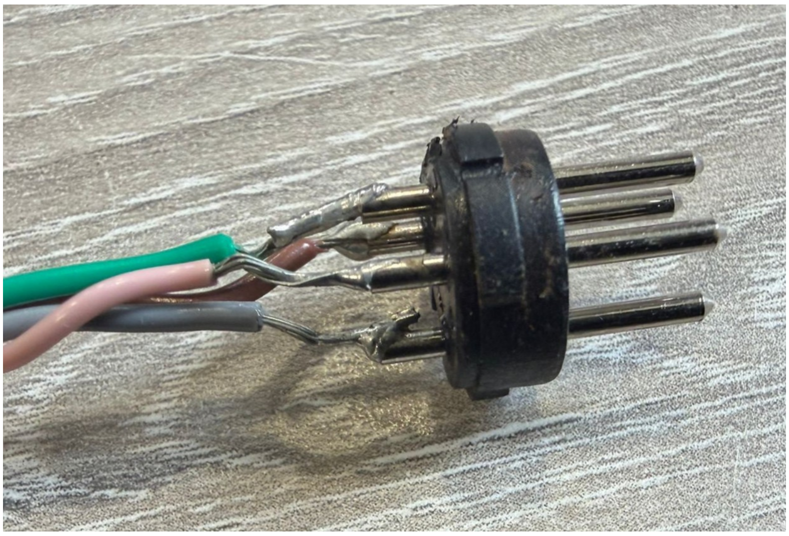


Figure 1: MALE DIN CONNECTOR WIRING

1.1 The Bus

1.1.1 Electrical Hardware

The Bus has an adaptor that supplies 5V to the device. Inside of the Bus, there is a perfboard that creates the 4 necessary lines for I2C communication: the Ground, Power, SDA and SCL.

The Ground and the Power can be directly sourced from the adaptor. However, Raspberry Pi Pico GPIO pins are not 5V tolerant, so the SCL and SDA wires need to have voltage of 3.3V. Currently this is accomplished by the use of another Raspberry Pi Pico that is powered by 5V and outputs 3.3V at its 3V3(OUT) pin. The SDA and SCL lines are connected to the Pico through individual 1kΩ pull-up resistors.

During testing, a 10kΩ pull-up resistor was not sufficient to pull the SDA and SCL above 3 V when other boxes were connected.

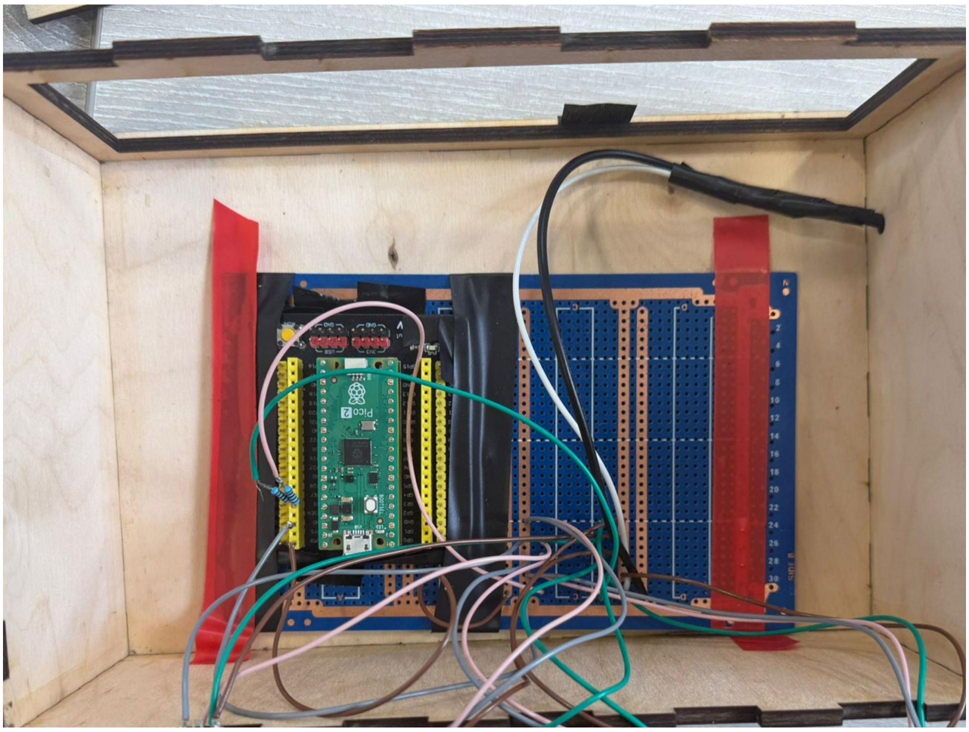


Figure 2: BUS BOX PERFBOARD

1.1.2 Top Face Layout

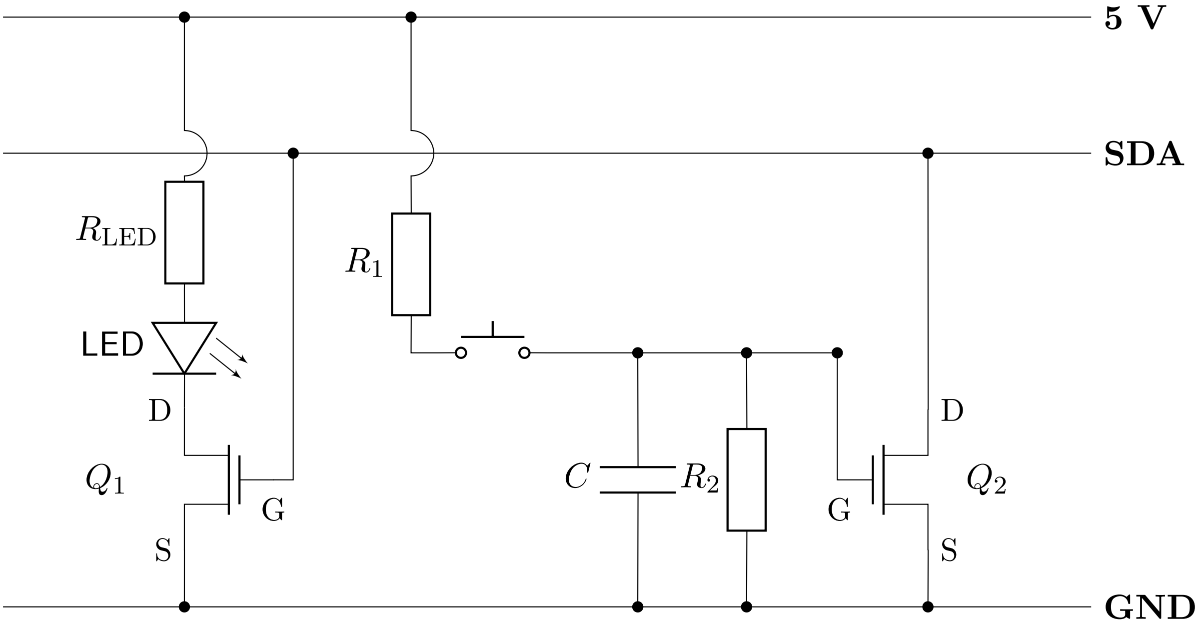
The top face layout of the Bus needs holes for the 4-pin DIN connectors. Every DIN jack can be fixed through screws(not implemented in the prototype).

Figure 3: BUS BOX TOP LAYOUT

1.2 The Controller

1.2.1 Electrical Hardware

The Controller has a red and a blue button on the top face that pull the SDA and SCL lines low when pressed. The circuit is purely electrical. A circuit diagram for one of the lines is shown below:



The MOSFET Q_1 is open only when the SDA line is high, lighting the LED. When the button is pressed, V_{GS} on Q_2 rises to 5V, short-circuiting the SDA line. The capacitor C and resistor R_2 form a low-pass filter to debounce the button. In the prototype, the following values are used:

$$R_1 = 100\Omega \quad ; \quad R_2 = 100k\Omega \quad ; \quad C = 100\text{nF} \quad ; \quad Q_1, Q_2 = \text{BS170 MOSFET}$$

The value of R_{LED} depends on the LED used. The circuit is identical for the SCL line. Importantly, a stronger capacitor can be used (330 nF) for stronger debouncing.

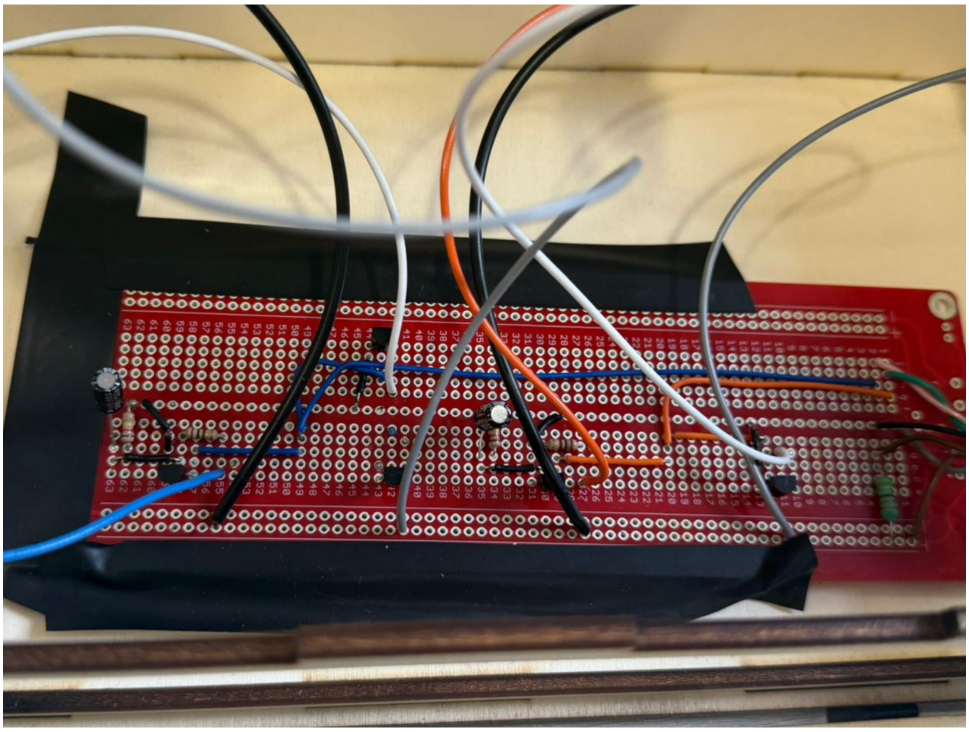


Figure 4: CONTROLLER PERFBOARD

1.2.2 Controller Top Face Layout

The top face currently has two holes through which the buttons are screwed and another for the ON LED. In the prototype, the LED indicators for the SCL and SDA line are located inside the button. For the final version, the LEDs will be located near the buttons, so more holes need to be added.

1.3 The Listener

1.3.1 Electrical Hardware

The Listener displays the state of the bits in the byte currently being transmitted, plus the ACK/NACK bit. For this, 9 LEDs are soldered on a perfboard and connected to a Raspberry Pi Pico. The 8 bit LEDs are blue, and the ACK LED is red. The LED connection to the Pico Pin is:

- BIT 7(MSB) - GP0
- BIT 6 - GP1
- BIT 5 - GP2
- BIT 4 - GP3
- BIT 3 - GP4
- BIT 2 - GP5

- BIT 1 - GP6
- BIT 0(LSB) - GP7
- ACK/NACK BIT - GP8

The SCL and SDA wires are connected to pins GP11 and GP10, respectively.

Note that the LEDs need to be soldered at a distance from the perfboard, to allow space for the resistors and wires when the LEDs pass through the holes in the top face of the box.

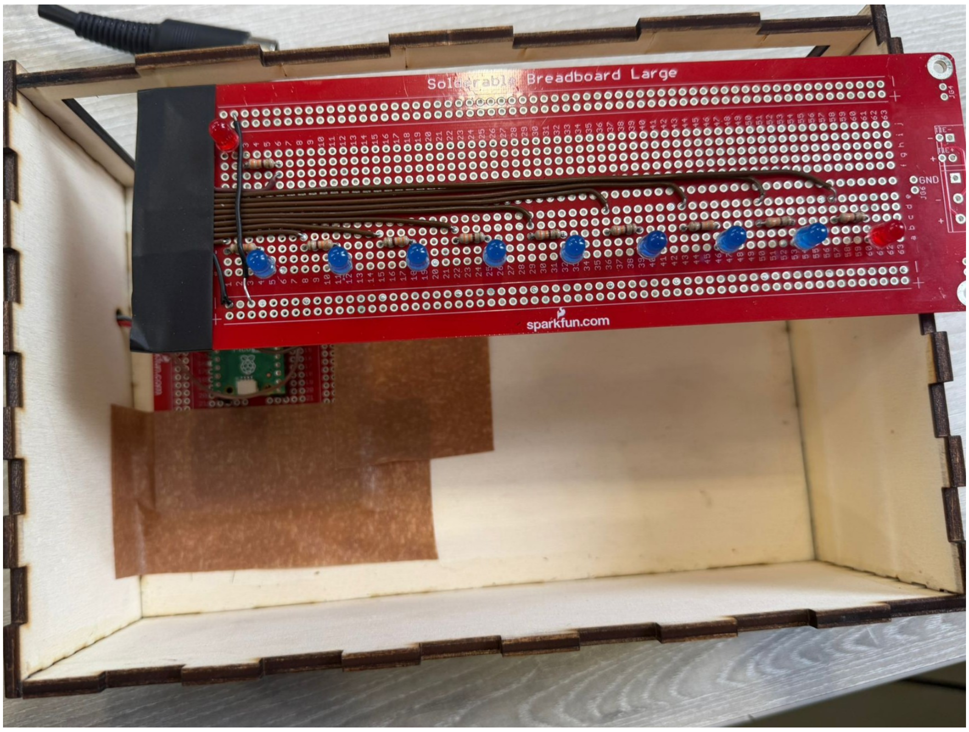


Figure 5: LISTENER PERFBOARD

1.3.2 Listener Top Face Layout

The top face of the Listener has a total of 10 LED holes: 8 for the byte, 1 for the ACK/NACK bit, and 1 for the ON LED.

1.4 The Target Box

1.4.1 Electrical Hardware

The Target box has 16 LEDs used to show information about the 4 simulated 'targets'. These are arranged in 4 rows and 4 columns. Every target can be in one of 4 states, represented by a column of the LED matrix:

- IDLE - the target is not listening to the I^2C bus;
- START - the target is recording the ADDRESS FRAME and checks that it matches its own;
- RECEIVING - the target is in receiving mode;
- TRANSMITTING - the target is in transmitting mode;

The connections between the LEDs and the Pico are:

- TARGET 1 - address 0x30

1. IDLE LED - GP2
2. START LED - GP3
3. RECEIVING LED - GP4
4. TRANSMITTING LED - GP5

- TARGET 2 - address 0x32
  1. IDLE LED - GP6
  2. START LED - GP7
  3. RECEIVING LED - GP8
  4. TRANSMITTING LED - GP9
- TARGET 3 - address 0x34
  1. IDLE LED - GP10
  2. START LED - GP11
  3. RECEIVING LED - GP12
  4. TRANSMITTING LED - GP13
- TARGET 4 - address 0x36
  1. IDLE LED - GP14
  2. START LED - GP15
  3. RECEIVING LED - GP16
  4. TRANSMITTING LED - GP17

The SCL and SDA lines are connected to pins GP19 and GP18, respectively. As with the Listener, the LEDs need to have space between them and the perfboard, to allow space for the resistor and wires.

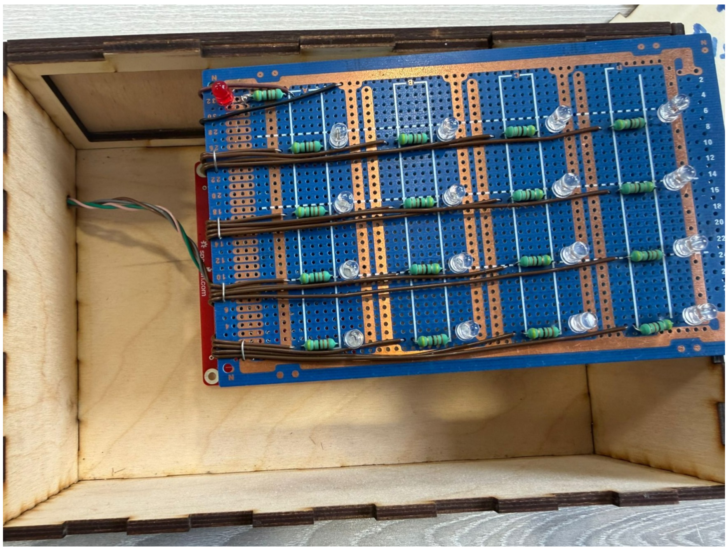


Figure 6: TARGET PERFBOARD

1.4.2 Target Top Face Layout

The Target has 17 LED holes on the top face: a matrix of 4x4 for the status LEDs and one more for the ON LED.